

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I harinesh.S 192521449 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

Question 1: Missing OSI Layers and Their Functions

Completed Concept Map

Application $\searrow$ Presentation $\searrow$ **Session** $\searrow$ Transport $\searrow$ Network $\searrow$ **Data Link** $\searrow$ Physical.

Functional Analysis

- **Session Layer:**
 - Establishes, manages, and terminates communication sessions between local and remote applications.
 - Handles authentication, authorization, and session restoration (checkpointing) so a connection can resume from its last known good state if interrupted.
 - **Data Link Layer:**
 - Provides reliable node-to-node data transfer across a physical link.
 - Packages raw bits from the Physical layer into structured frames.
 - Implements physical addressing (MAC addresses), flow control, and error detection to ensure local data delivery is accurate.
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Question 2: Data Units and OSI Layers Mapping

Concept Map

- **Transport Layer** $\searrow$ **Segment** (or Datagram for UDP)
- **Network Layer** $\searrow$ **Packet**
- **Data Link Layer** $\searrow$ **Frame**
- **Physical Layer** $\searrow$ **Bits**

Transformation Analysis (Encapsulation and Decapsulation)

- **Downstream (Encapsulation):** Data moves down the stack, and each layer adds structural control data (headers/trailers). The Transport layer splits data into manageable **segments**. The Network layer adds IP addresses to form **packets** for routing. The Data Link layer wraps packets into **frames** with physical MAC addresses. The Physical layer converts frames into electrical or optical **bits** for medium transmission.
 - **Upstream (Decapsulation):** The receiving device strips away these headers step-by-step to process and verify the data. This structured transformation abstracts complex networking tasks, letting each layer focus purely on its specific operational job.
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Question 3: Addressing and Delivery Systems

Analysis of Relationships

- **Data Link Layer $\rightarrow$ MAC Address $\rightarrow$ Local Delivery:** This relationship operates within a single local network segment (LAN). Devices use hardware-burned MAC addresses to forward data frames directly between adjacent nodes on the same physical link.
 - **Network Layer $\rightarrow$ IP Address $\rightarrow$ Global Delivery:** This relationship spans across multiple interconnected networks. Logical IP addresses allow routers to determine the best path across different subnets, moving packets from the original source network to the final destination network.
 - **Ensuring Complete Delivery:** Local delivery alone cannot cross a router, and global delivery alone does not know which specific machine on a local switch should receive the data. Together, IP routing gets the data to the correct neighborhood, and MAC addressing delivers it to the correct house.
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Question 4: Error Control Interaction Across Layers

Interaction Analysis

- **Data Link Layer (Error Detection):** This layer handles immediate, hop-by-hop integrity checks. It utilizes mathematical checks like Cyclic Redundancy Check (CRC) in the frame trailer to identify if bits were corrupted over the physical wire. If a frame is broken, it is typically discarded right away.
 - **Transport Layer (Error Correction):** This layer provides end-to-end reliability. It uses sequence numbers and acknowledgments (like TCP ACKs). If the receiving end notices a missing segment (potentially dropped by the Data Link layer below), it requests a retransmission from the original source.
 - **Reliability Improvement:** This split responsibility saves network bandwidth. Catching errors early at the Data Link layer prevents corrupted data from traveling further up the stack, while the Transport layer guarantees that any dropped data is successfully recovered.
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Question 5: Web Browsing Concept Map and Failure Analysis

Web Browsing OSI Concept Map

- **Application / Presentation / Session Layers:** HTTP/HTTPS protocols structure web requests; TLS/SSL handles encryption.
- **Transport Layer:** TCP establishes a reliable connection via a 3-way handshake to destination port 80 or 443.

- **Network Layer:** IP routes the web traffic across the internet using source and destination IP addresses.
- **Data Link / Physical Layers:** Ethernet or Wi-Fi transmits the web data as physical signals over local hardware.

Failure Cascading Analysis

- **The Impact:** If a lower layer fails (e.g., a broken Network Layer due to a lost IP route), the entire upper communication chain collapses.
 - **Why it happens:** Layers are strictly dependent on the services provided by the layers directly beneath them. If the Network layer cannot route packets, the Transport layer cannot maintain its TCP connection, preventing the Application layer from loading the website. The user sees a "Network Timeout" or "No Internet" error.
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Question 6: Troubleshooting via Concept Map

Problem Analysis

- **Physical (\(\checkmark\)) and Data Link (\(\checkmark\)):** Local hardware, cables, Wi-Fi connections, and local switch communications are functioning perfectly.
- **Network (\(\mathbf{X}\)):** The communication failure originates right here at the Network layer.
- **Transport (\(\mathbf{X}\)) and Application (\(\mathbf{X}\)):** These layers fail as a direct consequence of the Network layer failure.

How the Concept Map Helps in Troubleshooting

- **Isolates the Fault:** The map tells a technician to completely skip troubleshooting cables, network cards, or software applications.
 - **Targets the Root Cause:** It pinpoints the investigation directly to Network layer issues. A technician should immediately check for misconfigured static IP addresses, dead default gateways, subnets mismatches, or faulty router tables.
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RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation